

Adaptive Learning-driven MAC Protocol for Mobile Underwater Acoustic Sensor Networks

Slavica Tomović
Faculty of Electrical Engineering
University of Montenegro
Podgorica, Montenegro
slavicat@ucg.ac.me

Igor Radusinović
Faculty of Electrical Engineering
University of Montenegro
Podgorica, Montenegro
igorr@ucg.ac.me

Abstract— In this paper, we propose a Medium Access Control (MAC) protocol for mobile Underwater Acoustic Sensor Networks (UWASNs), based on framed-ALOHA and Reinforcement Learning (RL) principles. The protocol capitalizes on the large propagation delays in underwater acoustic channels to enhance network throughput. Deployed sensor nodes function as RL agents, autonomously learning optimal transmission strategies concerning both time slot allocation and packet transmission delays. This approach eliminates the need for gathering detailed information about the network topology and strengthens the network's resilience and adaptability. To evaluate its performance, we conducted UWASN simulations with different node mobility patterns. The simulation results show that the proposed protocol outperforms the existing RL-based UW-ALOHA-Q solution, achieving up to 50.7% improvement in channel utilization under the most challenging conditions, while sustaining a high level of fairness.

Keywords— AUV, MAC, mobility, reinforcement learning, underwater acoustic sensor networks.

I. INTRODUCTION

Mobile Underwater Acoustic Sensor Networks (UWASNs) have emerged as a cutting-edge solution for efficient data collection and monitoring in underwater environments. Including a mix of mobile nodes, which can range from Autonomous Underwater Vehicles (AUVs) and Unmanned Underwater Vehicles (UUVs) to free-floaters — these networks offer an expanded observational footprint and increased resilience against node failures [1-2].

In practical applications, mobile UWASNs can be swiftly deployed to specific underwater locations for short-term real-time missions. These missions encompass diverse objectives, including target detection and tracking, monitoring oil and chemical spills, and comprehensive underwater surveillance [1]. Within this operational framework, mobile nodes assume a pivotal role, continuously tracking local underwater activities and transmitting the sensed data over acoustic channel to enable collaborative data analysis among neighboring nodes or seamless consolidation of information at remote data collection centers. The collected data becomes a valuable resource that enhances decision-making capabilities.

The exciting potential of UWASNs comes with significant challenges in terms of communication protocol design. In this paper, we focus on Medium Access Control (MAC) protocols, tasked with coordinating packet transmissions among sensor nodes to reduce collision risks. The challenges in MAC protocol design arise due to unique properties of underwater acoustic channels: small propagation speed (~ 1500 m/s), narrow bandwidth, and high bit error rates. In such conditions,

where propagation delays typically far exceed packet transmission delays, traditional protocols tailored for terrestrial networks prove impractical and inefficient [1, 3]. Node mobility further complicates MAC design, requiring adaptations to cope with dynamic node locations and to ensure resilience to rapidly changing environmental conditions.

Over the past decade, MAC protocols for mobile UWASNs have attracted significant attention [1-6]. However, the majority of the proposed solutions rely on either carrier sensing or handshaking processes, both of which introduce substantial communication overhead [1]. Furthermore, most existing approaches mandate network-wide synchronization, which is hard to achieve due to GPS unavailability underwater [3]. An exception is APD-TDMA [6], designed for AUV networks. However, its primary concept revolves around estimating the future positions of AUVs based on packet arrival times at a central node. Consequently, this protocol is better suited for networks with nodes moving at a constant velocity, rather than those with variable speeds or directions.

In recent research, Reinforcement Learning (RL) has emerged as a promising tool for designing UWASN MAC protocols with low communication overhead [3, 7-9]. These protocols treat sensor nodes as RL agents, enabling them to learn good transmission strategies through interaction with their environment. The UW-ALOHA-Q protocol introduced in [3] demonstrates that RL can optimize time slot selection for collision-free access, eliminating the need for network-wide synchronization. Building upon this work, we proposed the DR-ALOHA-Q protocol [7] which empowers nodes to learn optimal policies for both time-slot and transmission-offset selection within a slot. This paper primarily focuses on an in-depth performance evaluation of DR-ALOHA-Q under dynamic mobile conditions, taking into account the impact of packet loss. While it is acknowledged that RL algorithms may face challenges in achieving convergence in highly dynamic settings, they exhibit the potential to adapt rapidly to environmental changes, significantly reducing the collision probability when compared to conventional approaches. We also broaden our study to encompass fairness metrics and propose modifications to DR-ALOHA-Q for better fairness outcomes. We compare this the proposed protocol, denoted as DR-ALOHA-QM, with UW-ALOHA-Q [3] and DOTS [1] in two distinct mobile UWASN scenarios. Simulation results demonstrate that DR-ALOHA-QM outperforms both in terms of channel utilization and achieves satisfying level of fairness.

The rest of the paper is organized as follows. In Section II we present the DR-ALOHA-QM protocol. Simulation results are presented in Section III. Section IV concludes the paper.

II. DR-ALOHA-QM PROTOCOL

In this section, we introduce the network model under consideration along with the DR-ALOHA-QM protocol. This protocol builds upon the DR-ALOHA-Q protocol, initially proposed in a prior work [7].

We consider UWASN model from [3], which involves N underwater sensor nodes that transmit sensed data to a central sink node. From the standpoint of the MAC protocol, time is organized into frames, each containing N time slots. At the beginning of every time frame, every node that selects one of the time slots for packet transmission. Additionally, nodes select a time-offset to control when their packet transmission occurs within the chosen time slot. Upon receiving a packet, the sink broadcasts an acknowledgment (ACK) packet over a dedicated downlink channel.

The optimization of sensor node transmission strategies, including time-slot and time-offset selection, is achieved through the implementation of an RL approach. In this context, each sensor node operates as an independent RL agent, undergoing training to make strategic decisions. The essence of RL lies in the interactive relationship between the agent and its environment. When these agents take actions, they receive feedback from the environment, manifesting as either rewards or penalties. In the specific scenario we are considering, rewards are earned for successful, collision-free packet transmissions, while penalties are incurred in the event of collisions. These rewards and penalties play a pivotal role in shaping the agent's decision-making policy.

In the proposed protocol design, the Q-learning RL approach is adopted. In Q-learning, the objective for the agent is to determine an optimal policy that maximizes the cumulative discounted reward [10]. The agent employs a Q-table, which serves as a knowledge base about the environment. This table contains rows corresponding to every conceivable environmental state and columns corresponding to feasible actions that the agent can take in each state. A value in the table, denoted as $Q(s_t, a_t)$, represents the expected reward achievable by taking action a_t in state s_t and subsequently adhering to the optimal policy from time t . The agent's policy is always greedy with respect to the Q-values. In the beginning, the Q-table is populated with random values. As the agent interacts with the environment, these values are adjusted according to the Bellman equation [10].

In the DR-ALOHA-QM protocol, the state is defined as the recipient of a data packet. Since the network model considered here involves only one node acting as a data receiver (the sink), the Q-learning method employed is a single-state Q-learning approach. This approach has been favored in recent studies ([3, 7, 9]) for designing MAC protocols due to its low computational complexity.

In this context, actions encompass decisions related to the transmission strategy, specifically involving the joint selection of a time-slot within a frame and a time-offset (delay) for packet transmission within that chosen time-slot. We quantify the time-offset options as multiples of δ time units, with a maximum allowable time-offset of $M\delta$ for packet transmission within each time slot. Therefore, for each sensor node i , the action space is defined as:

$$A_i = \{(j, m): 1 \leq j \leq N, 0 \leq m \leq M\} \quad (1)$$

where (j, m) represents the action of transmitting in the j -th slot of the subsequent frame, with a time delay of $m\delta$ units from the beginning of that slot.

A sensor node assesses the efficacy of its actions through the reward signal r_{t+1} , which is derived from feedback provided by the sink node. It awaits an acknowledgment packet (ACK) up to a specified deadline that is larger than the network's maximum round-trip time. If an ACK is received within this time window, the packet is deemed successfully received; otherwise, a packet collision is inferred. The reward varies based on the observed outcome (o_{t+1}) of the executed action. Receipt of an ACK always leads to a fixed, positive reward, whereas observed collisions result in a negative reward or penalty. DR-ALOHA-QM aim to optimize not only overall network throughput but also fairness among nodes. To this end, the penalty values are dynamically adjusted based on the throughput performance over a recent time window. Specifically, each node monitors its success rate over its last 50 transmission attempts (S_r) and uses this data to scale the penalty. The issue we intend to address is the inherent unfairness caused by differences in link propagation delays. Specifically, nodes that are closer to the sink node are in a more advantageous position for learning, as their interactions with the sink are faster. This could lead to an unfair distribution of network capacity among nodes. To mitigate this without incurring additional communication overhead, the reward function is redefined compared to [7] to incorporate historical throughput data:

$$r_{t+1} = \begin{cases} 1, & o_{t+1} = \langle \text{ACK} \rangle \\ -1 \cdot \min(1, 0.2 + \sqrt{S_r}), & o_{t+1} = \langle \text{Collision} \rangle \end{cases} \quad (2)$$

In each time frame, nodes with pending packets choose the action corresponding to the maximum Q-value. Given the state and action space definitions, each node's Q-table is one-dimensional, containing $N \cdot M$ elements. Upon observing a reward or penalty, the Q-value is updated as follows:

$$Q_{t+1}(a) = Q_t(a) + \gamma(r_{t+1} - Q_t(a)) \quad (3)$$

Here, a denotes the action taken, and γ is the learning rate. This update rule conforms to the standard Bellman equation for stateless Q-learning.

In UWASNs, due to large propagation delays, collision probability is influenced not only by packet transmission times but also by node locations - a phenomenon termed time-space uncertainty [1]. Traditional MAC approaches that extend time slots to accommodate maximum propagation delay T_{prop}^{max} and packet transmission time T_p sacrifice channel utilization. Alternatively, the DR-ALOHA-QM protocol leverages large propagation delays to permit parallel packet transmissions within the same contention domain. The protocol employs a reduced time slot size equal to $T_p + T_G$, where T_G is the guard interval. Previous work [7] demonstrated that a guard interval not exceeding 50% of T_p efficiently mitigates environmental variability and enables fast RL convergence in static UWASN settings.

Since the protocol operates under reduced time-slot constraints, ACK packets for successful transmissions may not always be received within the same time frame in which the action was taken. Also, ACK reception deadlines are calibrated to include both the maximum round-trip time and packet transmission delay, generally exceeding the frame duration prevalent in most network configurations. This

diverges from conventional RL models, where immediate feedback is expected prior to executing subsequent actions.

III. SIMULATIONS RESULTS

In this section, we assess the performance of the DR-ALOHA-QM protocol through simulations conducted using a Python-based, event-driven simulator. Simulations are carried out for UWASNs of varying sizes, ranging from 5 to 20 nodes, deployed over a 1km x 1km area. DR-ALOHA-QM is benchmarked against two existing protocols: UW-ALOHA-Q and DOTS. UW-ALOHA-Q employs RL exclusively for time-slot allocation and optimizes frame size by adjusting the number of slots. In this protocol, the slot size remains consistent with standard framed-ALOHA, encompassing maximum round-trip time, packet and ACK transmission durations, as well as a guard interval. On the other hand, DOTS capitalizes on passively gathered local data, such as propagation delay maps and expected transmission schedules from neighboring nodes, to maximize both temporal and spatial reuse. This approach enhances the likelihood of concurrent transmissions while reducing collision risks. To offer a comprehensive evaluation, DR-ALOHA-QM is tested under two distinct UWASN mobility scenarios: an AUV-assisted static network and a free-floating sensor network. The parameters common to all simulations are outlined in Table 1. Packet error rates are derived from results presented in [11] for acoustic channels with a strongly pronounced multipath.

TABLE I. SIMULATION PARAMETERS

Category	Value
Data packet size	1044 bits
ACK size	20 bits
Transmission rate	13.9 kbps
Propagation speed	1500 mps
γ	0.1
DR-ALOHA-QM slot size	110 ms
DR-ALOHA-QM δ	5 ms
Modem source level	160 dB re 1 μ Pa @1m
Modem bandwidth	8 kHz
Ambient noise spectral density	50 dB re μ Pa @1m /Hz
Absorption coefficient	6 dB/km

AUV-assisted UWASN includes static sensor nodes anchored on the seabed and a single AUV traversing a lawn-mower trajectory at speeds between 1 and 3 knots (0.51 mps to 1.54 mps). This network setup, with AUV as data mule, finds practical applications in oceanographic research, where AUVs can efficiently collect data from a network of fixed underwater sensors distributed across a wide area, significantly reducing the time and resources required for data retrieval and analysis. Conversely, static underwater nodes can provide critical support to more complex AUV missions by assisting in navigation, data collection, and mission planning, further enriching the capabilities of underwater exploration.

Fig. 1 compares channel utilization results of the analyzed protocols across varying UWASN sizes, under saturated traffic load. The results show that DR-ALOHA-QM significantly outperforms its counterparts. The DOTS protocol was notably less effective, especially in configurations with a small number of nodes. This can be attributed to the protocol's reliance on a handshaking mechanism, where the time required for channel reservation increases with the transmitter-receiver distance. Additionally, substantial propagation delays compromise the ability of

DOTS nodes to detect active transmissions effectively, leading to a high rate of collisions.

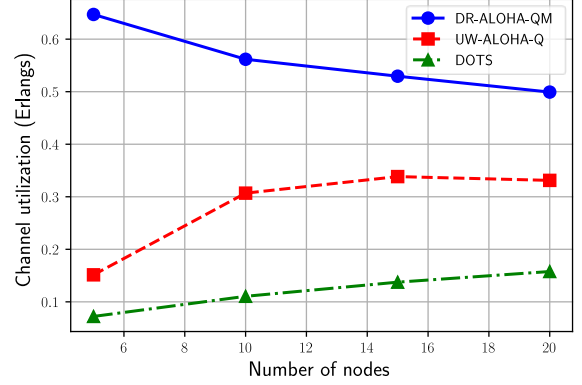


Fig. 1. Channel utilization in AUV-assisted UWASN under saturated traffic load.

The maximum channel utilization of UW-ALOHA-Q is limited by the number of slots in the frame, which is calculated based on the targeted index ratio ($B=1.5$) [3]. This ratio quantifies the relationship between the total time available for the sink node's reception within a frame and the cumulative duration of data packets produced by all nodes. In sparsely deployed UWASNs with small number of nodes, a frame size of just a single time slot is used, resulting in extended idle intervals within the frame and reduced channel utilization. While the large frame size can accommodate AUV mobility to large extent, the learning performance is severely degraded even by a modest number of consecutive collisions due to the adoption of random backoff strategy [3]. In contrast, DR-ALOHA-QM leverages RL to optimize both time-slot selection and time-offset adjustments for packet transmissions. This eliminates the need for random backoffs, resulting in increased learning stability. Despite the increased action space, DR-ALOHA-QM improves channel utilization by 50.7% relative to UW-ALOHA-Q in the least favorable conditions.

Fig. 2 presents channel utilization outcomes for a 10-node UWASN under diverse Poisson traffic loads. Notably, traffic irregularities in packet arrival times pose challenges for RL-based algorithms, as nodes often encounter contradictory feedback signals from the sink for identical actions. The results indicate a notable decline in DR-ALOHA-QM's performance under these conditions. However, it continues to surpass UW-ALOHA-Q in all but the two lowest traffic rates examined.

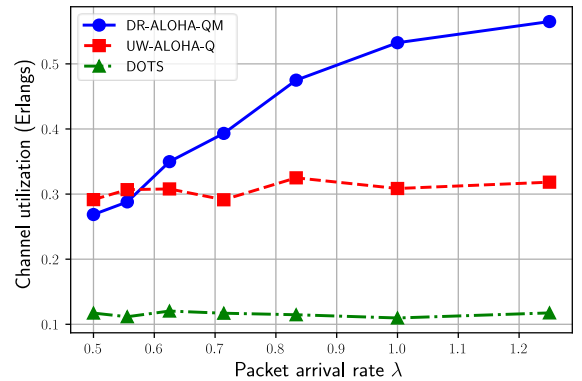


Fig. 2. Channel utilization in AUV-assisted UWASN under Poisson traffic load.

In addition to the AUV-assisted static UWASN setup, we also evaluated protocol performance in a UWASN configuration with free-floating sensor nodes. For this analysis, we posited a sink centrally within a 1km x 1km area, while the remaining nodes moved based on the Meandering Current Mobility (MCM) model [12], influenced by meandering subsurface currents and vortices. The MCM model parameters were configured as in [1, 3]. Fig. 3 shows the results for UWASNs with varying node counts. DOTS maintained performance levels comparable to those in the first scenario. However, UW-ALOHA-Q's performance deteriorated substantially in this setting, primarily due to its inadequate frame dimensioning approach and frequent triggering of its random backoff mechanism. DR-ALOHA-Q's performance was marginally better in this configuration, owing to the reduced maximum propagation time due to centrally located sink node.

Finally, in Fig. 4, we present fairness outcomes across both mobility models examined. For fairness evaluation, we employed the bottom-10% fair share (F_{10}) metric, which quantifies the ratio of the real bandwidth allocated to the least-performing 10% of nodes to the bandwidth they would receive under fair allocation (0-1 value). Owing to space constraints, we display results only for the most challenging conditions faced by DR-ALOHA-QM, specifically a 20-node UWASN configuration. The results suggest that DR-ALOHA-QM outperforms DOTS but lags behind UW-ALOHA-Q by 12.9% in the least favorable case. Despite this, the results are satisfying, given the substantial total throughput gains achieved by DR-ALOHA-QM.

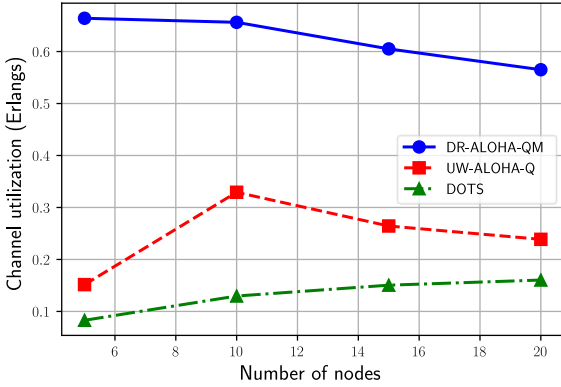


Fig. 3. Channel utilization in free-floating UWASN under saturated traffic load.

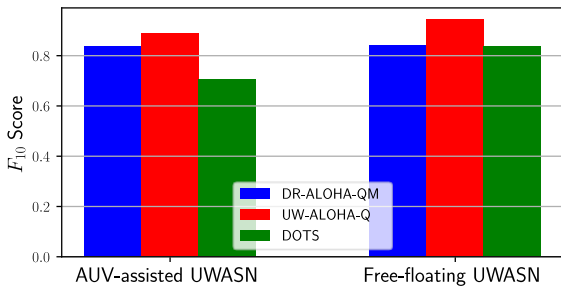


Fig. 4. F_{10} score in 20-node UWASN.

IV. CONCLUSION

In this paper, we introduced DR-ALOHA-QM, an RL-based MAC protocol tailored for mobile UWASNs where node movements are unpredictable. Built on the framed-ALOHA architecture, DR-ALOHA-QM employs RL to fine-tune time-slot assignments and time-offset adjustments for packet transmission, guided by sink node feedback. To encourage fair bandwidth distribution, a reward function is shaped to factor in the success of node's transmissions within a predefined time window. The protocol's performance was assessed in AUV-assisted and free-floating UWASN configurations. The obtained results show that DR-ALOHA-QM effectively adapts to varying conditions, with nodes learning sufficiently fast from their experiences. Notably, DR-ALOHA-Q outperformed the UW-ALOHA-Q protocol in terms of channel utilization, registering up to a 50.7% improvement in the most challenging scenarios, while also maintaining a high level of fairness.

In future work, we aim to refine DR-ALOHA-QM by incorporating adaptive frame size to better deal with traffic load fluctuations.

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